PRODUCT REQUIREMENT DOCUMENT

Operation Resilience: Corporate Risk & Supply Chain Intelligence Platform

Document ID	PRD-ENT-2026-V1	**Release Status**	Approved / Baseline
Project Title	Operation Resilience (OR-Platform)		
Author / Title	Principal PM, Supply Chain & Enterprise Risk	**Target Release**	Q3 2026 (MVP Phase 1)
Version	1.0	**Last Updated**	October 2026

1. EXECUTIVE SUMMARY

The geopolitical crisis and escalation of hostilities in the Middle East—specifically impacting the Strait of Hormuz—have severely compromised global shipping lanes, triggered immediate crude oil price volatility (+35%), and caused severe delays across cross-continental supply chains. Multinational enterprises face compounding operational risks: primary maritime routes are obstructed, sudden tariff changes are implemented unilaterally by sovereign nations, and tier-1/tier-2 suppliers face existential operational halts.

Operation Resilience is a enterprise-grade web application engineered to grant executive decision-makers end-to-end visibility, predictive risk scoring, and automated rerouting recommendations. By unifying real-time maritime telematics, geopolitical news synthesis, and internal ERP data, the platform empowers companies to transition from reactive crisis mitigation to proactive supply chain defense.

2. KEY PERFORMANCE INDICATORS (KPIS)

Mean Time to Detect (MTTD)

< 15 Min

Reduction from standard 48hr delay in disruption awareness.

Reroute Impact Efficiency

85%

Automated viable path recommendations within 1 hr.

Margin Margin Loss Saved

$12M+

Target annual savings per tier-1 enterprise subscriber.

Strategic Mandate

The primary imperative is establishing single-pane-of-glass transparency across ocean freight, energy exposure, and multi-tier vendor dependencies before supply chain bottlenecks immobilize core product assembly and customer delivery.

3. PROBLEM STATEMENT & DEEP ANALYSIS

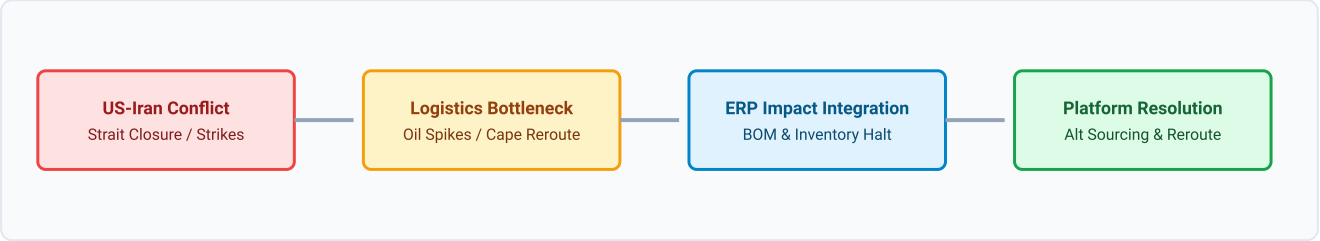
Multinational corporations are currently operating in a blind-spot environment due to legacy ERP systems relying on batch updates and static vendor lists. Modern enterprise risk management is broken along three critical dimensions:

- **Fragmented Intelligence:** Risk data is siloed across ocean carrier status logs, news alerts, and manual broker spreadsheets.
- **Delayed Response Vector:** Identifying that a shipment trapped near the Persian Gulf affects a secondary assembly plant in Europe currently takes 3 to 5 days.
- **Cost Volatility Invisibility:** Instantaneous spikes in war-risk insurance premiums and bunker fuel surcharges are missed during procurement planning.

4. TARGET AUDIENCE & DETAILED PERSONAS

Persona / Role	Primary Goals & Needs	Core Pain Points Solved
Chief Risk Officer (CRO) Executive Level	• Maintain enterprise resilience. • Quantify total exposure to board members. • Protect operational profitability.	Replaces reactive board reporting with real-time risk index scores and automated financial hazard models.
Supply Chain Director Operations Level	• Ensure transit continuity. • Manage ocean freight diversions. • Prevent factory downtime.	Eliminates blind spots on inbound vessel status, rerouting bottlenecks around choke points like Hormuz or Bab-el-Mandeb.
Procurement Manager Tactical Level	• Rapidly source secondary suppliers. • Control surging freight costs. • Monitor tier-2 vendor health.	Automates alternative supplier identification and calculates immediate cost/lead-time impact trade-offs.

OPERATIONAL CASCADE ARCHITECTURE



5. FUNCTIONAL REQUIREMENTS MATRIX

ID	Feature Module	Detailed Functionality & Logic	Priority	Complexity
FR-01	Live Conflict Geospatial Map	Interactive Mapbox overlay showing active kinetic zones, vessel positions (AIS data), naval blockades, missile range arcs, and port congestion indices in real-time.	P0	High
FR-02	Multi-Tier Dependency Graph	Automated ingestion of ERP Bill of Materials (BOM) mapped to vendor plant geographies down to Tier-3 suppliers. Visually flags single-source dependencies in high-risk zones.	P0	High
FR-03	Predictive Risk Scoring (PRS)	Machine learning algorithm generating dynamic 0-100 risk indexes per shipment based on route proximity to conflict, carrier risk profile, and commodity sensitivities.	P1	High
FR-04	Alternative Sourcing Engine	Recommends pre-vetted alternative suppliers outside conflict zones with side-by-side comparison of unit cost, lead time shift, and capacity availability.	P0	Med
FR-05	Real-Time Event Alerting Engine	Customizable push, email, and Webhook alerts triggered by specific geopolitical events (e.g., fuel price +10%, port closure, war insurance premium updates).	P1	Low
FR-06	Financial Exposure Modeler	Calculates immediate financial impact of delays including demurrage costs, contract default penalties, and expedited air-freight conversion options.	P2	Med

OPERATIONAL SCENARIO SIMULATION

> **Automated Resolution Workflow Example**
> **Trigger:** AIS telemetry confirms Port of Jebel Ali access restricted.
> **System Action:** Platform automatically tags 14 inbound containers containing critical automotive microchips. PRS score surges from 22 to 94.
> **Recommendation:** System prompts Logistics Lead to divert cargo to Salalah Port with air-bridge transfer to Frankfurt, providing a calculated cost differential of +$14,200 vs 18-day delay prevention.

6. NON-FUNCTIONAL REQUIREMENTS (NFRS)

Security & Compliance

SOC2 Type II certified, ISO27001 compliant. End-to-end encryption (AES-256) for stored ERP data and TLS 1.3 in transit.

Latency & SLA

Sub-2 second map rendering speed. Real-time AIS stream ingestion latency under 3 seconds. 99.95% system uptime SLA.

Scalability

Auto-scaling Kubernetes deployment supporting concurrent processing of 10M+ active shipment GPS coordinate pairs.

7. DATA STRATEGY & SYSTEM INTEGRATIONS

The architecture utilizes a unified Data Lakehouse model, combining continuous external streaming feeds with enterprise internal data pipelines.

Integration Category	Data Providers & Source	Protocol & Ingestion Strategy
Maritime Telematics	Spire Maritime, MarineTraffic API, AIS Hub	Kafka Stream / WebSocket continuous real-time vessel tracking feeds.
Geopolitical Intelligence	Janes Defense API, ACLED, Reuters/Bloomberg Feeds	REST API polling every 5 minutes with LLM NLP categorization.
Enterprise ERP / SCM	SAP S/4HANA, Oracle SCM, Kinaxis	Bi-directional GraphQL / REST endpoints for automated BOM ingestion.
Commodities & Energy	S&P Global Platts, ICE Brent Futures API	Webhooks triggered on >2% intraday price shifts.

Sign-off & Approval Gate

By signing below, leadership approves the functional scope and data architecture outlined in this PRD for MVP Phase 1 engineering execution.

Chief Risk Officer: _____ **VP of Engineering:** _____